Implementing a Ramsey Tax Plan in a Stochastic Continuous Time Economy*

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Abstract

We extend a deterministic continuous-time formulation of a Lucas and Stokey (1983) optimal tax problem to a stochastic setting driven by a Markov chain. To render markets dynamically complete, we equip the government with locally risk-free instantaneous debt and state-contingent hedging contracts. The government finances cumulative liabilities by rolling over instantaneous debt and managing a hedging portfolio, without rescheduling inherited term debts. A local commitment condition that preserves the purchasing power of instantaneous debt induces continuation governments to confirm the Ramsey plan. The value function is additively separable in a continuous state variable that equals the product of cumulative liabilities and the household's marginal utility of consumption. This state variable evolves continuously even when Markov transitions cause jumps in bond prices and marginal utilities.

Keywords: Ramsey planner, continuation Ramsey planner, implementability, instantaneous debt, local commitment, dynamic programming, hedging, complete markets.

JEL Classification: E62, H21, H63

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1 Introduction

Lucas and Stokey (1983) studied optimal fiscal policy in a dynamic, stochastic, complete-markets economy. A government finances an exogenous stochastic process for government expenditures $\{G_t\}$ and services an inherited term debt coupon process $\{b_{0-,t}\}$ by choosing a process of distorting flat rate taxes. Lucas and Stokey characterized a Ramsey plan that maximizes the representative household's welfare and showed how to implement it by appropriately rescheduling the term structure of government debt, thereby inducing a sequence of continuation governments to confirm the Ramsey tax plan. Debortoli, Nunes, and Yared (2021) showed that Lucas and Stokey's implementation can fail when initial debt is so high that the Ramsey planner sets the tax rate above the peak of the Laffer curve.

Jiang et al. (2024), a non-stochastic predecessor of this paper, showed that this difficulty can be resolved by expanding the government's contractible subspace to include "instantaneous debt" — a locally risk-free security of infinitesimal maturity — together with a local commitment condition that requires each continuation government to preserve the "purchasing power" of outstanding instantaneous debt, measured as the product of the debt balance and the representative household's marginal utility of consumption. The deterministic setting of Jiang et al. (2024) featured government expenditure and initial term debt service processes that were known functions of calendar time, so that bond prices were known at time 0 and no hedging was needed.

This paper extends Jiang et al. (2024) to a stochastic setting while remaining faithful to Lucas and Stokey's (1983) specification of complete markets. In continuous time, this requires some care. Lucas and Stokey's (1983) original discrete-time complete markets assumption, namely that Arrow securities for every state and date are traded at time 0, has a well-known continuous-time counterpart in finance: the dynamic spanning constructions of Black and Scholes (1973), Merton (1973), and Harrison and Kreps (1979) that use instantaneous debt together with competitively priced hedging contracts that pay off at state transitions. Our stochastic environment features a time-homogeneous Markov chain $\{s_t\}$ with n states; starting from state s_{t-}, there are $n-1$ possible transitions, each associated with a hedging contract. Together, the instantaneous (locally risk-free) debt and these $n-1$ hedging contracts span all contingencies over each infinitesimal interval, rendering markets dynamically complete. In contrast with the deterministic predecessor, the government must actively manage both instantaneous debt and a portfolio of hedging contracts

in order to finance its cumulative liabilities under the Ramsey plan. Our main results carry over from the deterministic setting, but the proofs are more involved because state transitions induce jumps in bond prices, marginal utilities, and the cumulative liability process.

Section 2 describes the environment: a continuous-time economy with a representative household, a benevolent government, and Markovian exogenous processes $\{G_t,\, b_{0-,t}\}$. The household's first-order conditions pin down the stochastic discount factor and establish a key implementability constraint that encodes the government's present-value budget constraint purely in terms of allocations.

Section 3 formulates a Lagrangian for the Ramsey problem, attaching a multiplier Λ_0 to the implementability constraint. Under the Markov assumption, optimal consumption $C(s_t; \Lambda_0)$ depends only on the current state and the multiplier. This section also derives the equilibrium risk-free interest rate, which equals the household's subjective discount rate adjusted for risk premia associated with state transitions, and the market price of state-transition risk.

Section 4 reformulates the Ramsey problem as a dynamic program. We define the cumulative government liability $\Pi_t = \int_0^t \frac{q_{0,u}}{q_{0,t}} (b_{0-,u} + G_u - \tau_{0,u} N_{0,u})\, du$, which tracks what the government owes absent any rescheduling of its initial term debt. While Π_t can jump when the Markov state transitions, the product $X_t = \Pi_t\, U_{C,t}$ — the "purchasing power" of cumulative liabilities — evolves continuously in time. This continuity makes X_t a natural state variable. We show that the value function takes the additively separable form $V(X_t, s_t) = -\Phi_0\, X_t + H(s_t;\, \Phi_0)$, where the scalar Φ_0 is chosen by the Ramsey planner to maximize $H(s_0;\, \Phi_0)$ subject to the implementability constraint. The Lagrangian multiplier Λ_0^* from section 3 equals Φ_0^*, establishing that the Lagrangian and Bellman equation approaches yield the same Ramsey plan.

Section 5 shows how the government finances its cumulative liabilities. We extend Lucas and Stokey's complete markets assumption to our continuous-time stochastic setting by allowing the government to trade locally risk-free instantaneous debt and $n-1$ competitively priced hedging contracts. The government finances the Ramsey plan by leaving the initial term debt structure $\{b_{0-,t}\}$ untouched and adjusting only its instantaneous debt balance and hedging positions. The instantaneous debt balance B_t then equals the cumulative liability Π_t^*, while the hedging demand $\Delta_t^{(s_{t-},j)} = (1 - e^{-\eta^*(s_{t-},j)})\Pi_{t-}^*$ offsets state-transition risk.

Section 6 addresses the question that motivated Lucas and Stokey (1983) and Debortoli,

Nunes, and Yared (2021): how to implement the Ramsey plan sequentially under a timing protocol that grants "partial commitment" to continuation governments. We adopt a local commitment condition: each government's current-period tax rate is set by its immediate predecessor. Under this timing protocol, the continuation government at time t inherits an instantaneous debt balance $B_t = \Pi_t^*$ and the tail of the original term debt structure, and faces the same purchasing-power-adjusted state variable $X_t^* = B_t U_{C,t}^*$ as the initial Ramsey planner. We show that the continuation government then voluntarily confirms the Ramsey plan.

Section 7 presents quantitative illustrations in a two-state regime-switching environment. When the initial term debt process $\{b_{0-,t}\}$ switches between high and low levels while government spending is constant, the Ramsey planner sets taxes low and interest rates low in the high-debt state, manipulating bond prices to reduce the market value of its debt. The cumulative liability Π_t^* jumps at state transitions, but $X_t = \Pi_t U_{C,t}$ remains continuous. When instead government spending $\{G_t\}$ switches while the initial debt service is constant, the Ramsey planner smooths taxes completely, holding consumption, the tax rate, and the interest rate all constant — an echo of Barro (1979)'s finding that intertemporal properties of an optimal tax rate can be disconnected from those of the government expenditure process.

Section 8 constructs an alternative recursive formulation based on a forward-looking promised liability Ω_t that represents the present discounted value of all future government liabilities. The associated state variable $Y_t = \Omega_t U_{C,t}$ has a law of motion whose coefficient on the martingale increment is indeterminate, reflecting the government's freedom in choosing Arrow securities. Despite this indeterminacy, the value function retains the same additively separable structure and the Ramsey plan is unique: the optimal Ψ_0^* equals Φ_0^*, so consumption and tax paths are identical to those derived in section 4. Section 9 offers concluding remarks.

2 Environment

Time $t \in [0, \infty)$ is continuous. A representative household and benevolent government participate in a complete set of perfectly competitive markets. The government finances a stochastic exogenous stream of expenditures with a stream of distorting flat tax rates. The exogenous shock in the economy is driven by a time-homogeneous (continuous-time) Markov chain $\{s_t; t \geqslant 0\}$ where s_t takes values in the set $\mathbf{N} = \{1, 2, \cdots, n\}$. The probability

of s_t transitioning from $j \in \mathbf{N}$ to $k \in \mathbf{N}$ over a small time interval Δt is $\lambda_{j,k}\Delta t$ where $\lambda_{j,k} \geqslant 0$ for $k \neq j$. The probability of remaining in j over Δt is then $\big(1 - \lambda_{j,j}\Delta t\big)$, where $\lambda_{j,j} = -\sum_{k \neq j} \lambda_{j,k} \leqslant 0$. At time 0, a Ramsey planner selects the best competitive equilibrium with distorting taxes.

The representative household supplies a labor process $\{N_{0,t}; t \geqslant 0\}$ that produces a flow of a single nonstorable good that can be divided between a consumption flow $\{C_{0,t}; t \geqslant 0\}$ and a government expenditure flow $\{G_{0,t}; t \geqslant 0\}$:

$$C_{0,t} + G_t = N_{0,t}\,, \text{ for all } t \geqslant 0\,. \tag{1}$$

At time 0, the representative household orders consumption and labor supply streams $\{C_{0,t}; t \geqslant 0\}$ and $\{N_{0,t}; t \geqslant 0\}$ according to

$$\mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} U(C_{0,t}, N_{0,t}) dt \right], \tag{2}$$

where $\rho > 0$ and $U(\cdot, \cdot)$ is strictly increasing in consumption C, strictly decreasing in labor supply N, globally concave, and continuously differentiable. Along with Debortoli, Nunes, and Yared (2021), we assume that labor supply $N_{0,t}$ has no upper bound.

Let $\tau_{0,t}$ denote a tax rate at t chosen at time 0 and $q_{0,t}$ be the time-0 value of a zero-coupon bond with a unit payoff at t. A first subscript denotes the time that a variable is chosen and that a second subscript denotes a time that the variable is realized.

The representative household faces a single intertemporal budget constraint:

$$\mathbb{E}_0 \left[\int_0^\infty C_{0,t} q_{0,t}\, dt \right] \leqslant \mathbb{E}_0 \left[\int_0^\infty b_{0-,t} q_{0,t} dt \right] + \mathbb{E}_0 \left[\int_0^\infty (1 - \tau_{0,t}) N_{0,t} q_{0,t} dt \right]. \tag{3}$$

The left side of (3) is the present value of the household's consumption stream $\{C_{0,t}; t \geqslant 0\}$ and the right side is the sum of the household's financial wealth and its human wealth in the form of the present value of after-tax labor income. Given tax rate process $\{\tau_{0,t}; t \geqslant 0\}$ and bond price process $\{q_{0,t}; t \geqslant 0\}$, the household chooses $\{C_{0,t}, N_{0,t}; t \geqslant 0\}$ to maximize (2) subject to constraint (3). That optimum problem yields the following first-order necessary conditions:

$$1 - \tau_{0,t} \;=\; -\frac{U_{N,t}}{U_{C,t}}\,, \tag{4}$$

$$q_{0,t} \quad = \quad e^{-\rho t}\frac{U_{C,t}}{U_{C,0}}\,. \tag{5}$$

The tax rate $\tau_{0,t}$ and the stochastic discount factor (SDF) $q_{0,t}$ are both stochastic processes.

A time-0 government finances an initial debt and its exogenous spending stream by taxing labor income:

$$\mathbb{E}_0\left[\int_0^\infty b_{0-,t}q_{0,t}dt\right] \leqslant \mathbb{E}_0\left[\int_0^\infty q_{0,t}\left(\tau_{0,t}N_{0,t}-G_t\right)dt\right], \tag{6}$$

The left side is time 0 value of payouts on government debt: $\{b_{0,t}; t \geqslant 0\}$ and the right side is time 0 value of $\{S_{0,t}; t \geqslant 0\}$. We let $S_{0,t} = \tau_{0,t}N_{0,t} - G_t$ denote the primary surplus.

Definition 1. *Given the government's initial debt term structure $\{b_{0-,t};\, t \geqslant 0\}$, spending flow process $\{G_t;\, t \geqslant 0\}$, and a flat rate tax process $\{\tau_{0,t};\, t \geqslant 0\}$, a* ***competitive equilibrium*** *is a feasible allocation, $\{C_{0,t}, N_{0,t};\, t \geqslant 0\}$, and a bond price process $\{q_{0,t};\, t \geqslant 0\}$ for which*

- *the government's budget constraint* (6) *is satisfied and*
- *the allocation $\{C_{0,t}, N_{0,t};\, t \geqslant 0\}$ solves the household's optimization problem.*

We follow Lucas and Stokey (1983) and use first-order conditions (4) and (5) together with feasibility constraint (1) to eliminate tax rates and bond prices from the government budget constraint. We thereby obtain the following implementability constraint on competitive equilibrium allocations $\{C_{0,t}, N_{0,t}\}_{t=0}^{\infty}$:

$$\mathbb{E}_0\left[\int_0^\infty e^{-\rho t}(C_{0,t}U_{C,t} + (C_{0,t}+G_t)U_{N,t})dt\right] \geqslant \mathbb{E}_0\left[\int_0^\infty e^{-\rho t}b_{0-,t}U_{C,t}dt\right]. \tag{7}$$

Proposition 1. *Given an initial term-debt structure $\{b_{0-,t}; t \geqslant 0\}$ and government spending path $\{G_t; t \geqslant 0\}$, an allocation $\{C_{0,t}, N_{0,t}; t \geqslant 0\}$ is a competitive equilibrium allocation if and only if it satisfies (1) for all $t \geqslant 0$ and (7) at $t = 0$.*

Ramsey problem. A Ramsey planner chooses a process $\{C_{0,t}; t \geqslant 0\}$ that maximizes

$$\mathbb{E}_0\left[\int_0^\infty e^{-\rho t}U(C_{0,t}, C_{0,t}+G_t)dt\right] \tag{8}$$

subject to implementability constraint (7). By using first-order conditions (4) and (5), we can compute an associated tax rate process and bond price system. After the Ramsey planner chooses a tax plan and associated price system, the representative agent's best responses confirm the associated competitive equilibrium allocation.

To ease exposition, for the remainder of the paper we follow Lucas and Stokey (1983) by focusing on a Markovian setting:

Assumption 1. *Exogenous flows of government expenditures* $\{G_t; t \geqslant 0\}$ *and debt-service payouts* $\{b_{0,t}; t \geqslant 0\}$ *are described by the following (time-homogenous) functions of the state variable* s_t*:*

$$G_t = G(s_t) \quad \textit{and} \quad b_{0-,t} = b(s_t)\,. \tag{9}$$

3 A Lagrangian

In this section, we construct a Lagrangian for the Ramsey problem, use it to characterize the equilibrium allocation associated with the Ramsey plan, then derive equilibrium asset-pricing implications.

3.1 Posing the Ramsey Problem

Attach a multiplier Λ_0 to implementability constraint (7) and form a Lagrangian:

$$\mathcal{L}_0 = \mathbb{E}_0\left[\int_0^\infty e^{-\rho t}\left[U(C_{0,t}, C_{0,t} + G(s_t)) + \Lambda_0\left(C_{0,t}U_{C,t} + (C_{0,t} + G(s_t))U_{N,t} - b(s_t)U_{C,t}\right)\right]dt\right]. \tag{10}$$

The Ramsey planner maximizes the right side of (10) over consumption plans $\{C_{0,t}\}_{t=0}^\infty$ and minimizes over the nonnegative multiplier Λ_0. We denote the extremizing values $\{C^*_{0,t}\}_{t=0}^\infty$ and Λ_0^*. For a given Λ_0, the optimal consumption plan satisfies [1]

$$C(s_t; \Lambda_0) := \arg\max_{C_{0,t}}\left[U(C_{0,t}, C_{0,t} + G(s_t)) + \Lambda_0\left(C_{0,t}U_{C,t} + (C_{0,t} + G(s_t))U_{N,t} - b(s_t)U_{C,t}\right)\right]. \tag{11}$$

[1] We assume a regularity condition under which the optimal consumption plan for the problem defined in (11) is unique for a given Λ_0.

Substituting (11) into implementability constraint (7), we deduce that Λ_0 must satisfy:

$$\mathbb{E}_0\left[\int_0^\infty e^{-\rho t}\big[\,(C(s_t;\Lambda_0)-b(s_t))\,U_{C,t}+(C(s_t;\Lambda_0)+G(s_t))U_{N,t}\big]dt\right]=0\,. \tag{12}$$

Substituting $C_{0,t} = C(s_t;\Lambda_0)$ into the right side of (10) lets us write $\mathcal{L}_0$ as a function $\mathcal{L}_0(\Lambda_0)$. An optimal Λ_0^* is the non-negative root of equation (12) that maximizes $\mathcal{L}_0(\Lambda_0)$. The Ramsey allocation satisfies $C_{0,t}^* = C(s_t;\Lambda_0^*)$ and $N_{0,t}^* = C_{0,t}^* + G_t = N(s_t;\Lambda_0^*)$.

Proposition 2. *Under conditions provided in Appendix A, there exists a non-negative root Λ_0^* of equation (12) that solves*

$$\Lambda_0^* = \arg\min_{\Lambda_0}\ \mathcal{L}_0(\Lambda_0)\,.$$

The Ramsey allocation, tax rate, and bond price process satisfy

$$C_{0,t}^* = C(s_t;\Lambda_0^*);\quad N_{0,t}^* = C_{0,t}^* + G(s_t);$$
$$\tau_{0,t}^* = 1+\tfrac{U_{N,t}(C_{0,t}^*,N_{0,t}^*)}{U_{C,t}(C_{0,t}^*,N_{0,t}^*)};\quad q_{0,t}^* = e^{-\rho t}\tfrac{U_{C,t}(C_{0,t}^*,N_{0,t}^*)}{U_{C,0}(C_{0,0}^*,N_{0,0}^*)}\,.$$

The value $\mathcal{L}_0^*$ of a Ramsey plan is $\mathbb{E}_0\big[\int_0^\infty e^{-\rho t}U(C_{0,t}^*, C_{0,t}^*+G(s_t))dt\big]$.

3.2 Asset Prices

Applying Ito's Lemma to $q_{0,t}^*$ given in (5), we obtain

$$dq_{0,t}^* = -\rho q_{0,t-}^* dt + q_{0,t-}^*\frac{dU_{C,t}}{U_{C,t-}}\,, \tag{13}$$

where the last term captures the effect of state transitions on the $\{q_{0,t}\}$ process. The equilibrium SDF in a market economy where shocks are driven by jumps can be written as (see, e.g., Duffie (2001)):

$$dq_{0,t}^* = -r_{0,t-}^* q_{0,t-}^* dt - q_{0,t-}\sum_{j\neq s_{t-}}\zeta^*(s_{t-},j)\big(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j}dt\big)\,, \tag{14}$$

where $r_{0,t-}^*$ is the equilibrium interest rate and $\zeta^*(s_{t-},j)$ is the equilibrium market price of risk associated with a transition to $j \neq s_{t-}$ from s_{t-}. Equilibrium requires that the expected rate of change of $q_{0,t}^*$ equals $-r_{0,t-}^*$. Using the martingale representation theorem for a jump

process, we thus can write the $q_{0,t}$ process as (14), where the conditional expectation of the second term at $t-$ equals zero.

Both (13) and (14) must hold, so using restrictions that equilibrium puts on the drift of the SDF and the jump size from state $s_{t-} = i$ to each state $j \neq i$, we obtain:

$$\zeta^*(i,j) = -\left(e^{\eta^*(i,j)} - 1\right), \tag{15}$$

where $\eta^*(i,j)$ for $i \neq j$ is

$$\eta^*(i,j) = \ln \frac{U_C^*(C(j), N(j))}{U_C^*(C(i), N(i))} \tag{16}$$

and the equilibrium risk-free rate is

$$r_{0,t-}^* = r^*(s_{t-}) = \rho - \sum_{j \neq s_{t-}} \left(e^{\eta^*(s_{t-},j)} - 1\right) \lambda_{s_{t-},j}\,. \tag{17}$$

Equation (16) implies that the market price of risk associated with transition from state i to j is positive, i.e., $\zeta^*(i,j) > 0$, if $\eta^*(i,j) < 0$. This requires that the marginal utility of consumption decreases when the economy transitions to state j from state i. Thus, if consumption increases when the state transitions from $s_{t-} = i$ to $s_t = j$, the zero-coupon bond price decreases from $q_{0,t-}$ to $q_{0,t} = q_{0,t-} e^{\eta^*(i,j)}$ where $j \neq i$, as we can see from the jump shock piece in (14) and by noting $d\mathcal{J}_t^{(s_{t-},j)} = 1$. Equation (17) characterizes the equilibrium risk-free rate: the second term captures both expected growth and precautionary savings effects that are associated with a jump in the equilibrium interest rate.

4 A Dynamic Program

Via the following four steps, we the Ramsey problem recursively: 1. For a given consumption process, we define a process, $\{\Pi_t;\, t \geqslant 0\}$ that tracks the government's cumulative liabilities in the absence of any debt restructuring. 2. We use the $\{\Pi_t;\, t \geqslant 0\}$ process to construct a state variable process $\{X_t;\, t \geqslant 0\}$. 3. We frame a dynamic program with $\{X_t;\, t \geqslant 0\}$ as a state variable and a state s_t dependent value function, $V(X_t, s_t)$ that is additively separable: $V(X_t, s_t) = -\Phi_0\, X_t + H(s_t;\, \Phi_0)$, where Φ_0 is an undetermined scalar. 4. We compute Φ_0. The dynamic program sets the stage for a state-contingent debt management strategy that implements the Ramsey plan.

4.1 Cumulative Liabilities with no Debt Restructuring

Given a competitive equilibrium, we construct

$$\Pi_t = \int_0^t \frac{q_{0,u}}{q_{0,t}} \left(b_{0-,u} + G_u - \tau_{0,u} N_{0,u}\right) du \,, \quad \Pi_0 = 0 \,. \tag{18}$$

Evidently, the ratio $q_{0,u}/q_{0,t}$ compounds the contribution of the flow $(b_{0-,u}+G_u-\tau_{0,u}\, N_{0,u})\, du$ to the government's time-t liability stock.[2] The $\{\Pi_t\}$ process tracks the cumulative liabilities with no debt restructuring at all from 0 to t.

Lemma 1. Π_t *defined in* (18) *evolves according to*

$$\begin{aligned} d\Pi_t \quad &= \quad r_{0,t-}\Pi_{t-}dt + \left(b_{0-,t-} + G_{t-} - \tau_{0,t-}N_{0,t-}\right) dt \\ &\quad + \sum_{j \neq s_{t-}} \left(e^{-\eta(s_{t-},j)} - 1\right) \Pi_{t-} \left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} e^{\eta(s_{t-},j)} dt\right) , \end{aligned} \tag{19}$$

where $\eta(s_{t-}, s_t) = \ln \frac{U_{C,t}}{U_{C,t-}}$.

In evolution equation (19), $\lambda_{s_{t-},j} e^{\eta(s_{t-},j)}$ in the last term is the 'risk-adjusted' transition rate[3] at which the state transitions from s_{t-} to $j \neq s_{t-}$ under the physical measure. The ratio between the risk-adjusted and original transition rates, $e^{\eta(s_{t-},j)}$, captures the 'price' of transition risk from state s_{t-} to j. The representative consumer demands a positive risk premium when a state transition increases the marginal utility of consumption: $\eta(s_{t-}, j) > 0$. When the state transitions from s_{t-} to $j \neq s_{t-}$, i.e., when $d\mathcal{J}_t^{(s_{t-},j)} = 1$, the cumulative liability jumps from Π_{t-} to $\Pi_{t-} e^{-\eta(s_{t-},j)} = \Pi_{t-} \frac{U_{C,t-}}{U_{C,t}} = \Pi_{t-} \frac{q_{0,t-}}{q_{0,t}} = \Pi_t$. Consequently, while Π_t and $q_{0,t}$ can both jump, their product $q_{0,t}\Pi_t$ is continuous at all t.

4.2 A State Variable X_t

The continuity of the $\{q_{0,t}\Pi_t\}$ process induces us to define

$$X_t = \Pi_t U_{C,t} \,. \tag{20}$$

[2] When the interest rate is a constant r over (u, t), $q_{0,u}/q_{0,t} = e^{r(t-u)} > 1$.

[3] Technically, $\lambda_{s_{t-},j} e^{\eta(s_{t-},j)}$ is the regime-transition rate under the risk-neutral measure; see, e.g., Duffie (2001), for a textbook treatment.

Evidently, X_t incorporates both information about Π_t and the marginal utility of consumption $U_{C,t}$. Combining (20), (18) and (5) yields

$$X_t = \Pi_t e^{\rho t} q_{0,t} U_{C,0} = e^{\rho t} U_{C,0} \int_0^t q_{0,u} \left(b_{0-,u} + G_u - \tau_{0,u} N_{0,u} \right) du \,. \tag{21}$$

Differentiating both sides of the above equation and using $C_t = N_t - G_t$, $e^{\rho t} U_{C,0} q_{0,t} = U_{C,t}$, and $(1 - \tau_{0,t}) U_{C,t} = -U_{N,t}$, implied by (1) and (4)-(5), we have

$$\begin{aligned} dX_t &= \rho X_t dt + e^{\rho t} U_{C,0} q_{0,t} \left(b_{0-,t} + G_t - \tau_{0,t} N_{0,t} \right) dt \\ &= \rho X_t dt + U_{C,t} \left(b_{0-,t} + G_t - N_{0,t} + (1 - \tau_{0,t}) N_{0,t} \right) dt \\ &= \left(\rho X_t + b_{0-,t} U_{C,t} - C_{0,t} U_{C,t} - N_{0,t} U_{N,t} \right) dt \,. \end{aligned} \tag{22}$$

We have established:

Proposition 3. *For a given competitive equilibrium, X_t defined in (20) is continuous in time and for all $t \geqslant 0$ evolves according to (22).*

4.3 Ramsey Problem as a Dynamic Program

We reformulate the Ramsey problem recursively in four steps. First, for a given X_0, we define and solve a dynamic programming (DP) problem with X_t and state s_t being the state variables in subsection 4.3. Then, in subsection 4.4 we use the DP problem to characterize the Ramsey plan.

Definition 2 (DP Problem). *Confronting X_0 at $t = 0$, the Ramsey planner solves*

$$\max_{\{C_{0,t}; t \geqslant 0\}} \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} U(C_{0,t}, C_{0,t} + G_t) dt \right], \tag{23}$$

where maximization is subject to the evolution equation (22).

Let $V(X_0, s_0)$ denote the (optimal) value function for this problem. Let

$$V(X_t, s_t) = \max_{\{C_{0,u}; u \geqslant t\}} \mathbb{E}_t \left[\int_t^\infty e^{-\rho(u-t)} U(C_{0,u}, C_{0,u} + G_u) du \right] \tag{24}$$

where maximization is subject to (22). Appendix A verifies that $V(X_t, s_t)$ satisfies the

following HJB equation:

$$\rho V = \max_{C_{0,t}} \; U(C_{0,t}, C_{0,t} + G_t) + \frac{\partial V}{\partial X_t}\left[\rho X_t - C_{0,t}U_{C,t} - (C_{0,t} + G_t)U_{N,t} + b_{0-,t}U_{C,t}\right] + \sum_{k \neq s_t} \lambda_{s_t,k}\left[V(X,k) - V(X,s_t)\right] . \tag{25}$$

Now guess and verify that $V(X_t, s_t)$ takes the following form:

$$V(X_t, s_t) = -\Phi_0 X_t + H(s_t; \Phi_0)\,, \tag{26}$$

where the scalar Φ_0 is to be chosen by the Ramsey planner. Substituting (26) into (25), we obtain the following interconnected system of equations for $H(s_t; \Phi_0)$ for each $s_t = 1, \cdots, n$:

$$\rho H(s_t; \Phi_0) = \max_{C_{0,t}} U(C_{0,t}, C_{0,t} + G_t) + \Phi_0\big[C_{0,t}U_{C,t} + (C_{0,t} + G_t)U_{N,t} - b_{0-,t}U_{C,t}\big] + \sum_{k \neq s_t} \lambda_{s_t,k}\left[H(k;\Phi_0) - H(s_t;\Phi_0)\right]. \tag{27}$$

Define

$$C(s_t; \Phi_0) = \arg\max_{C_{0,t}} U(C_{0,t}, C_{0,t} + G_t) + \Phi_0\left[C_{0,t}U_{C,t} + (C_{0,t} + G_t)U_{N,t} - b_{0-,t}U_{C,t}\right] \tag{28}$$

and let

$$N(s; \Phi_0) \;=\; C(s; \Phi_0) + G(s). \tag{29}$$

According to (28)-29, consumption $C_{0,t}$ and labor $N_{0,t}$ depend only on the contemporaneous exogenous state s_t and not on X_t.

$$\begin{aligned} \mathbb{V} &= [V(X,1;\Phi_0), \cdots, V(X,n;\Phi_0)]^{\intercal} \\ \mathbb{H} &= [H(1;\Phi_0), \cdots, H(n;\Phi_0)]^{\intercal} \; . \end{aligned}$$

Proposition 4. *For $s \in \{1, 2, \cdots, n\}$, let*

$$A(s;\Phi_0) = U(C(s;\Phi_0), C(s;\Phi_0) + G(s)) + \Phi_0\big[C(s;\Phi_0)U_C + (C(s;\Phi_0) + G(s))U_N - b(s)U_C\big].$$

Let

$$\begin{aligned}\mathbb{V} &= [V(X,1;\Phi_0),\cdots,V(X,n;\Phi_0)]^\intercal \\ \mathbb{H} &= [H(1;\Phi_0),\cdots,H(n;\Phi_0)]^\intercal \\ \mathbb{A} &= [A(1;\Phi_0),\cdots,A(n;\Phi_0)]^\intercal\ .\end{aligned}$$

Choices of $\{C_{0,t}\}$ *for the* **DP Problem** *indexed by* $\Phi_0 \geqslant 0$ *satisfy (28) and attain value function* (26)*, where* $H(s_t;\Phi_0)$ *is given by*

$$\mathbb{H} = (\rho\mathbb{I} - \Lambda)^{-1}\,\mathbb{A}\,, \tag{30}$$

4.4 Characterizing the Ramsey Plan

Setting $X_0 = 0$ and $\Pi_0 = 0$ and keeping the initial term-debt structure $\{b_{0-,t} = b(s_t); t \geqslant 0\}$ fixed, the expected utility attained by the representative agent under the Ramsey plan at state s_0 is

$$W(s_0) \;=\; \max_{\Phi_0}\; V(X_0, s_0; \Phi_0)\,, \tag{31}$$

where the maximization on the right side is subject to implementability constraint (7). Define

$$I_0(s_0;\Phi_0) = \mathbb{E}_0\left[\int_0^\infty e^{-\rho t}\left[C(s_t;\Phi_0)U_{C,t} + (C(s_t;\Phi_0) + G(s_t))U_{N,t} - b(s_t)U_{C,t}\right]dt\right]. \tag{32}$$

implementability constraint (7) requires that Φ_0 satisfies $I_0(s_0;\Phi_0) = 0$. For a given initial state s_0, let $\mathcal{A} = \{\Phi_0 \geqslant 0 : I_0(s_0;\Phi_0) = 0\}$ denote an admissible set for Φ_0. Using (26) and $X_0 = 0$, the representative agent attains utility criterion

$$W(s_0) \;=\; \max_{\Phi_0\in\mathcal{A}} V(0, s_0) \;=\; \max_{\Phi_0\in\mathcal{A}} H(s_0;\Phi_0)\,. \tag{33}$$

Note that Φ_0 does not appear explicitly in (33), since $X_0 = 0$ because $\Pi_0 = 0$; nevertheless the first term in (26), Φ_0 appears implicitly in (33) via the $H(s_0;\Phi_0)$ function shown in (30). Evidently, Φ_0 depends on s_0. We are now equipped to state:

Proposition 5. *When it exists, a Ramsey plan satisfies*

$$C^*_{0,t} = C(s_t; \Phi^*_0); \quad N^*_{0,t} = C^*_{0,t} + G(s_t);$$
$$\tau^*_{0,t} = 1 + \frac{U_{N,t}(C^*_{0,t}, N^*_{0,t})}{U_{C,t}(C^*_{0,t}, N^*_{0,t})}; \quad q^*_{0,t} = e^{-\rho t} \frac{U_{C,t}(C^*_{0,t}, N^*_{0,t})}{U_{C,0}(C^*_{0,0}, N^*_{0,0})}$$

where $\Phi^*_0 = \arg\max_{\Phi_0 \in \mathcal{A}} V(0, s_0)$. *Under a Ramsey plan, the representative agent's value is*

$$W(s_0) = H(s_0; \Phi^*_0). \tag{34}$$

5 Financing Deficits

We alter Lucas and Stokey's formulation by allowing the government to trade "instantaneous debt", a security typically included in complete markets models in finance, e.g., Black and Scholes (1973), Merton (1973), and Harrison and Kreps (1979). In this way, we extend Lucas and Stokey's assumption of complete markets to our continuous time setting.

5.1 Liability Management

We seek a dynamic portfolio management strategy that finances the government's cumulative liabilities, Π^*_t, under the Ramsey plan. Evaluating Π_t given in (19) under the Ramsey plan yields:

$$\begin{aligned} d\Pi^*_t - r^*_{0,t-}\Pi^*_{t-}dt = &\left(b_{0-,t-} + G_{t-} - \tau_{0,t-}N^*_{0,t-}\right) dt \\ &+ \sum_{j \neq s_{t-}} \left(e^{-\eta^*(s_{t-},j)} - 1\right) \Pi^*_{t-} \left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)} dt\right). \end{aligned} \tag{35}$$

We require a set of financial claims that allow us to finance $d\Pi^*_t - r^*_{0,t-}\Pi^*_{t-}dt$, the net change of Π^*_t over a small dt interval. The government can issue instantaneous debt that is locally risk free in the sense that over an infinitesimal time interval dt it returns $r^*_{0,t}dt$. Such instantaneous debt is not state-contingent. The government also has access to competitively priced hedging contracts that allow it to manage state-transition risk exposure. Because s_t resides in one of n states, starting from s_{t-}, we have $n-1$ distinct state transitions. For each possible transition from s_{t-} to $j \neq s_{t-}$, there is a hedging contract. Over a small time interval dt, the holder of a unit of such a hedging contract receives a unit payoff at t if and only if the state transitions from s_{t-} to another state $j \neq s_{t-}$ at time t, i.e., $d\mathcal{J}_t^{(s_{t-},j)} = 1$.

The holder pays an actuarially fair hedging premium, $\lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)}$, per unit of time before a transition occurs. The premium equals the transition rate $\lambda_{s_{t-},j}$ multiplied by a factor $e^{\eta^*(s_{t-},j)}$ that compensates for state-transition risk. If the transition to state j increases the marginal utility U_C, $\eta^*(s_{t-}, j) > 0$, then investors receive a positive hedging premium: note that $\lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)}$ exceeds state transition probability $\lambda_{s_{t-},j}$.

Together with the state non-contingent instantaneous debt, this collection of $(n-1)$ distinct hedging contracts spans all possible contingencies over a small time interval dt, so markets are dynamically complete in the tradition of Black and Scholes (1973), Merton (1973), and Harrison and Kreps (1979). Let $B_t^{(0)}$ denote the balance of the state non-contingent instantaneous debt at t. Let $\Delta_{t-}^{(s_{t-},j)}$ denote the hedging demand for state transition from s_{t-} to $j \neq s_{t-}$. Let $\partial_t b_{t-,u}$ denote the (incremental) term-debt coupon issued at time $t-$ and due at u.

The evolution of sources and uses of government funds satisfies

$$d\Pi_t^* - r_{0,t-}^* \Pi_{t-}^* dt = \underbrace{dB_t^{(0)} - r_{0,t-}^* B_{t-}^{(0)} dt}_{\text{net non-contingent instantaneous debt issuance}} + \underbrace{\left(\mathbb{E}_{t-} \int_{u>t} q_{t-,u}^* \partial_t b_{t-,u} du\right) dt}_{\text{gross state-contingent term-debt issuance}}$$
$$- \underbrace{\left(\int_{u<t} \partial_u b_{u,t-} du\right) dt}_{\text{incremental term debt due}} - \underbrace{\sum_{j \neq s_{t-}} \Delta_{t-}^{(s_{t-},j)} \left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)} dt\right)}_{\text{stochastic hedging payoff netting premium payments}}, \tag{36}$$

where Π_t^* defined in (18) is the cumulative liability under the Ramsey plan absent term debt adjustments. The first term on the right side of (36) describes the net instantaneous non-contingent debt issuances. The second term is gross state-contingent term-debt issuance over dt. The third term is the sum of maturing term debts that had been issued cumulatively from 0 to t. The last term equals the sum of contingent payoffs from all its $(n-1)$ hedging contracts, $\sum_{j \neq s_{t-}} \Delta_{t-}^{(s_{t-},j)} d\mathcal{J}_t^{(s_{t-},j)}$, minus the sum of hedging premium payments for all these contracts: $\sum_{j \neq s_{t-}} \Delta_{t-}^{(s_{t-},j)} \lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)} dt$. As the government's cumulative liability Π_t^* increases with the hedging premium payments and decreases by $\Delta_{t-}^{(s_{t-},j)}$ when a state transition from s_{t-} to $j \neq s_{t-}$ occurs, i.e., when $d\mathcal{J}_t^{(s_{t-},j)} = 1$, there is a minus sign in front of this underbraced term. Note that over a dt interval, either zero or one state transition occurs.

Equating the right sides of (35) and (36) yields:

$$\begin{aligned}
&\left(b_{0-,t-} + G_{t-} - \tau_{0,t-} N^*_{0,t-}\right) dt + \sum_{j \neq s_{t-}} \left(e^{-\eta^*(s_{t-},j)} - 1\right) \Pi^*_{t-} \left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)} dt\right) \\
= \quad & dB_t^{(0)} - r^*_{0,t-} B^{(0)}_{t-} dt + \left(\mathbb{E}_{t-} \int_{u>t} q^*_{t-,u} \partial_t b_{t-,u} du\right) dt - \left(\int_{u<t} \partial_u b_{u,t-} du\right) dt \\
& - \sum_{j \neq s_{t-}} \Delta_{t-}^{(s_{t-},j)} \left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)} dt\right) \qquad (37)
\end{aligned}$$

Since (37) holds at all t, matching the term for the transition from the current state s_{t-} to $j \neq s_{t-}$ yields the following hedging demand $\Delta_{t-}^{(s_{t-},j)}$:

$$\Delta_{t-}^{(s_{t-},j)} = \left(1 - e^{-\eta^*(s_{t-},j)}\right) \Pi^*_{t-} \,. \qquad (38)$$

Evidently, the government uses the Delta hedging presented by Black and Scholes (1973) and Merton (1973), appropriately adapted to our state-transition setting. Note that the hedging demand associated with a transition from s_{t-} to $j \neq s_{t-}$ is unique. Cancelling the last terms on both sides of (37), we obtain the following accounting identity:

$$\begin{aligned}
&\underbrace{(G_{t-} - \tau^*_{0,t-} N^*_{0,t-}) dt}_{\text{primary deficits}} + \underbrace{\left(b_{0-,t} + \int_{u<t} \partial_u \, b_{u,t-} \, du\right) dt}_{\text{all maturing term debt repayments}} \\
&\qquad = \underbrace{dB_t^{(0)} - r^*_{0,t-} B^{(0)}_{t-} dt}_{\text{net instantaneous-debt issuance}} + \underbrace{\left(\mathbb{E}_{t-} \int_{u>t} q^*_{t-,u} \, \partial_t \, b_{t-,u} \, du\right) dt}_{\text{gross term-debt issuance}} \,. \qquad (39)
\end{aligned}$$

5.2 No rescheduling of initial term debts

We describe a dynamic portfolio strategy that finances the government's cumulative liability $\{\Pi^*_t\}$ process by automatically adjusting the instantaneous state non-contingent debt balance $B_t^{(0)}$ and the $(n-1)$ state-contingent hedging demand functions $\Delta_{t-}^{(s_{t-},j)}$ for $j \neq s_{t-}$ while leaving all exiting term debts untouched and issuing no new term debts. Under this strategy, $\partial_t \, b_{t,u} = 0$ for all $(t \geqslant 0, u > t)$ and $t \geqslant 0$. Simplifying (39), we obtain the following process for the balance of the instantaneous state non-contingent debt $B_t^{(0)}$:

$$dB_t^{(0)} \quad = \quad r^*_{0,t-} B^{(0)}_{t-} dt + \left(b_{0,t-} + G_{t-} - \tau^*_{0,t-} N^*_{0,t-}\right) dt \,, \quad \text{with} \;\; B_0 = \Pi_0 = 0 \,. \qquad (40)$$

A hedging account tracks the cumulative losses due to hedging. Let $B_t^{(1)}$ denote the balance of this account starting with $B_0^{(1)} = 0$. If $B_t^{(1)} > 0$, the government has a cumulative loss at t. If $B_t^{(1)} < 0$, the government has a cumulative hedging gain at t. As this account tracks the cumulative hedging losses under the optimal hedging policy, the government can roll over the entire (loss) balance of this hedging account at the locally risk-free rate $r^*_{0,t-}$. This policy implies the following account balance dynamics:

$$dB_t^{(1)} = r^*_{0,t-} B_{t-}^{(1)} dt - \sum_{j \neq s_{t-}} \Delta_{t-}^{(s_{t-},j)} \left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)} dt \right). \tag{41}$$

where $\Delta_{t-}^{(s_{t-},j)}$ is given in (38).

Let $B_t = B_t^{(0)} + B_t^{(1)}$ denote the combined balances in the two accounts. Forward induction from $t = 0$ path by path implies $B_t = \Pi_t^*$ for all t. Thus, combining (40) with (42) and using (38), we obtain

$$\begin{aligned} dB_t &= r^*_{0,t-} B_{t-} dt + \left(b_{0,t-} + G_{t-} - \tau^*_{0,t-} N^*_{0,t-} \right) dt \\ &\quad + \sum_{j \neq s_{t-}} \left(e^{-\eta^*(s_{t-},j)} - 1 \right) B_{t-} \left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)} dt \right). \end{aligned} \tag{42}$$

Note that B_t incorporates the cumulative effects of the government's primary deficits and hedging losses (gains). Consequently, B_t perfectly tracks Π_t^*.

Thus, our dynamic portfolio strategy for financing Π_t^* over time is composed of the instantaneous state non-contingent debt $B_t^{(0)}$ given in (40) and the $(n-1)$ state-contingent hedging demand functions, $\Delta^{(s_{t-},j)}$ for $j \neq s_{t-}$ given in (38).

By contrast, the debt-management strategy of Lucas and Stokey (1983) uses active management of state-contingent term debts over time.

Remark 2. *The debt rescheduling policies of Lucas and Stokey (1983) use only term debt. A coupon flow management policy of* $\{\partial_t b_{t-,u}; u \geqslant t, t \geqslant 0\}$ *and* $\{B_t^{(0)} = 0; t \geqslant 0\}$ *is used to satisfy the dynamic budget constraint:*

$$\begin{aligned} d\Pi_t^* - r^*_{0,t-} \Pi^*_{t-} dt &= \left(\mathbb{E}_{t-} \int_{u>t} q^*_{t-,u} \partial_t b_{t-,u} du \right) dt - \left(\int_{u<t} \partial_u b_{u,t-} du \right) dt \\ &\quad + \sum_{j \neq s_{t-}} \left(e^{-\eta^*(s_{t-},j)} - 1 \right) \Pi^*_{t-} \left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} e^{\eta^*(s_{t-},j)} dt \right). \end{aligned} \tag{43}$$

This policy requires continuous adjustments to the term debt structure. That makes it

challenging to formulate the Ramsey problem recursively.

Relative to Lucas and Stokey (1983), our dynamic portfolio strategy has two key features: (1) distinguishing sharply between instantaneous debt and term debts, and (2) a menu of state-contingent hedging contracts involving minimal number of securities in the spirit of one-period Arrow securities (Arrow, 1964).

6 Implementing a Ramsey Plan

We now answer a question that concerned Lucas and Stokey (1983) and Debortoli, Nunes, and Yared (2021): how to implement the Ramsey plan sequentially under a timing protocol that attributes "partial commitment" to a sequence of government decision makers.[4] We adopt a timing protocol that adds a speck of partial commitment to a continuation Ramsey planner's choice of a tax policy. This lets us implement Ramsey plans that Lucas and Stokey can not.

6.1 Partial Commitment and Implementation

In order to induce an immediate successor government to confirm a current government's planned tax, we infinitesimally constrain Lucas and Stokey's (1983) governments in the following way. Let $\tau_{t-,t}$ denote a time-t tax rate applied over the interval $(t, t+dt)$ that is preferred by the government in office for the interval $(t-dt, t)$, which we index as the time $(t-)$ government.

Assumption 2 (Instantaneous partial commitment)**.** *A time-$t-$ government sets a time-t government's current period tax rate $\tau_{t,t}$:*

$$\tau_{t,t} = \tau_{t-,t}. \tag{44}$$

Thus, the time t tax rate that applies to taxable income over a short time interval $(t, t+dt)$ is set by the instantaneous predecessor of the time-t government, also known as the continuation planner at time $t-$. Via the household's first-order necessary condition, this timing protocol induces

$$C_{t,t} = C_{t-,t}, \tag{45}$$

[4]The concluding section of Lucas and Stokey (1982), the original working paper version of Lucas and Stokey (1983), contains a fascinating discussion of "partial commitment" and the way it shaped monetary-fiscal policy decision making arrangements in the U.S. during the first Washington administration.

where $C_{t-,t}$ is the time-t consumption effectively chosen by the time-$t-$ government.

To achieve this outcome, the government at time $t-$ hands over to the time-t government, a portfolio of financial assets that delivers a balance of instantaneous debt B_t for each possible state s_t at t and a stream of term debt services $\{b_{t-,u};\, u \geqslant t\}$. Importantly, we show that there exists a financial portfolio with the following properties for each s_t that always implements the Ramsey plan, provided that it exists:

$$B_t = \Pi_t^* \quad \text{and} \quad b_{t-,u} = b_{0-,u}; \quad u \geqslant t, \tag{46}$$

where Π_t^* is the state-contingent cumulative liability whose evolution is given in (35) under the Ramsey plan. To be precise, using the dynamic trading strategy that features a combination of state non-contingent debt and a portfolio of Arrow securities described in Subsection 5.1, we can show that (46) holds for all paths. As (46) requires that the term debt service flow inherited by the time-t government equals the tail $\{b_{0-,u};\, u \geqslant t\}$ of the initial term debt services stream at all t, the government's instantaneous debt balance B_t must perfectly track the government's cumulative liability Π_t^* for all paths. This is achieved by properly choosing the dynamically changing hedging contract holdings Δ_t given in (38).

Under Assumption 2 that specifies an instantaneous commitment, the dynamic trading strategy described above allows us to ensure that X_t defined in (20) equals X_t^* under the Ramsey plan. That is, the following identity holds:

$$B_t U_{C,t}(C_{t,t}, N_{t,t}) = \Pi_t^* U_{C,t}(C_{t-,t}, N_{t-,t}) = X_t^*, \tag{47}$$

where $C_{t,t}$ and $N_{t,t}$ are the time-t government's choices. Note that if the Ramsey plan has been implemented until time $t-$, then $C_{t-,t} = C_{0,t}^*$ and $N_{t-,t} = N_{0,t}^*$. Now we are ready to state our implementation result and then prove it.

Proposition 6. *Under our Assumption 2 timing protocol, a Ramsey plan can be implemented by accepting the tail of initial debt service stream* $\{b_{0-,u}\}$ *and adjusting instantaneous debt according to* (46) *.*

Proof. We use a recursion to prove the above proposition. Under the assumption that the Ramsey plan is implemented up to time $t-$, it is sufficient if we prove that the Ramsey plan is also implemented at time t. We can show that at time t the continuation Ramsey planner, confronting $B_t = \Pi_t^*$ and $b_{t-,u} = b_{0,u}$ for all $u \geqslant t$ and the inherited hedging contract holdings $\{\alpha(s_{t-}, j); j \neq s_{t-}\}$, will confirm the continuation of the Ramsey plan by

choosing $C_{t,u} = C^*_{0,u}$ and $N_{t,u} = N^*_{0,u}$ for all $u \geqslant t$. We can prove the preceding claim by subdividing time-t continuation Ramsey planner's problem into two subproblems.

First, the time-t continuation Ramsey planner chooses $\{C_{t,u}, u \geqslant t\}$ to attain

$$\max_{\{C_{t,u};u\geqslant t\}} \mathbb{E}_t \left[\int_t^\infty e^{-\rho(u-t)} U(C_{t,u}, C_{t,u} + G_u) du \right], \tag{48}$$

subject to $\widehat{X}_t = B_t U_{C,t}(C_{t,t}, N_{t,t})$ and the following dynamics for $u \geqslant t$:

$$d\widehat{X}_u \quad = \quad \big(\rho \widehat{X}_{u-} - C_{t,u-} U_{C,u-} - N_{t,u-} U_{N,u-} + b_{0,u-} U_{C,u-}\big) du\,. \tag{49}$$

Let $\widehat{V}(\widehat{X}_t, s_t; \Phi_t)$ denote the value function for the continuation planner defined in (48). Similar to our analysis in Section 4, we can show that $\widehat{V}(\widehat{X}_t, s_t; \Phi_t)$ is additively separable: $\widehat{V}(\widehat{X}_t, s_t; \Phi_t) = -\Phi_t \widehat{X}_t + \widehat{H}(s_t; \Phi_t)$, where $\widehat{H}(s_u; \Phi_t)$ for $u \geqslant t$ solves the following differential equation:

$$\begin{aligned} \rho \widehat{H}(s_u; \Phi_t) = \max_{C_{t,u}} & \; U(C_{t,u}, C_{t,u} + G_u) + \Phi_t \big[C_{t,u} U_{C,u} + (C_{t,u} + G_u) U_{N,u} - b_{0-,u} U_{C,u} \big] \\ & + \sum_{j \neq s_u} \lambda_{s_u,j} \left[\widehat{H}(j; \Phi_t) - \widehat{H}(s_u; \Phi_t) \right], \end{aligned} \tag{50}$$

where $C_{t,u} = C(s_u; \Phi_t)$, and $C(s_u; \Phi_t)$ is the same as the function defined in (28).

Under the local commitment condition given in Assumption 2, the time t continuation Ramsey planner has to choose $C_{t,t} = C_{t-,t} = C^*_{0,t}$. Because 1.) $B_t = \Pi^*_t$ under our implementation and 2.) $\widehat{X}_t = B_t U_{C,t}(C_{t,t}, N_{t,t}) = \Pi^*_t U_{C,t}(C^*_{0,t}, N^*_{0,t}) = X^*_t$, we have $\widehat{X}_t = X^*_t$.

We now turn to establishing that $\Phi^*_t = \Phi^*_0$. We start by proving that the time-t continuation Ramsey planner chooses $C_{t,u} = C^*_{0,u}$ for all $u > t$. Because the optimal continuation plan $\{C^*_{t,u};\, u > t\}$ satisfies $C^*_{t,u} = C(s_u; \Phi^*_t)$ for some $\Phi^*_t > 0$ and the Ramsey plan satisfies $C^*_{0,u} = C(s_u; \Phi^*_0)$, we need to verify that $\Phi^*_t = \Phi^*_0$, which we do via contradiction. Suppose that there exists an $\Phi^*_t \neq \Phi^*_0$ such that the associated continuation Ramsey plan yields a higher value than the tail of the Ramsey plan, meaning that

$$\mathbb{E}_t \int_t^\infty e^{-\rho(u-t)} U(C^*_{t,u}, C^*_{t,u} + G_u)\, du > \mathbb{E}_t \int_t^\infty e^{-\rho(u-t)} U(C^*_{0,u}, C^*_{0,u} + G_u)\, du, \tag{51}$$

where $C^*_{t,u} = C(s_u; \Phi^*_t)$. Recall that the time-$t$ continuation Ramsey planner chooses an

allocation $\{C^*_{t,u};\, u > t\}$ that satisfies the following continuation implementability condition:

$$\mathbb{E}_t \int_t^\infty e^{-\rho(u-t)} \big(C^*_{t,u}\, U_{C,u} + (C^*_{t,u} + G_u)\, U_{N,u}\big)\, du \geqslant \mathbb{E}_t \int_t^\infty e^{-\rho(u-t)} b_{0-,u}\, U_{C,u}\, ds + B_t\, U_{C,t}\,. \quad (52)$$

Construct a plan $\{\widetilde{C}_{0,s};\, s \geqslant 0\}$ in which the allocation from 0 to t follows the Ramsey plan, $\widetilde{C}_{0,s} = C^*_{0,s}$ for $s \in [0,t]$, but the allocation from time t onward follows the time-t continuation Ramsey plan: $\widetilde{C}_{0,s} = C^*_{t,s}$ for $s \in (t,\infty)$. Using (52) and $B_t = \Pi^*_t$, we can verify that $\{\widetilde{C}_{0,s};\, s \geqslant 0\}$ satisfies the implementability condition (7) for the Ramsey plan. Next, using (51), we can show that this proposed plan $\{\widetilde{C}_{0,s};\, s \geqslant 0\}$ yields a value for the household that is higher than the value under the Ramsey plan:

$$\begin{aligned}
&\mathbb{E}_0 \int_0^\infty e^{-\rho u} U(\widetilde{C}_{0,u}, \widetilde{C}_{0,u} + G_u)\, du \\
&= \mathbb{E}_0 \left[\int_0^t e^{-\rho u} U(\widetilde{C}_{0,u}, \widetilde{C}_{0,u} + G_u)\, du + \mathbb{E}_t \int_t^\infty e^{-\rho u} U(\widetilde{C}_{0,u}, \widetilde{C}_{0,u} + G_u)\, du \right] \\
&= \mathbb{E}_0 \left[\int_0^t e^{-\rho u} U(C^*_{0,u}, C^*_{0,u} + G_u)\, du + \mathbb{E}_t \int_t^\infty e^{-\rho u} U(C^*_{t,u}, C^*_{t,u} + G_u)\, du \right] \\
&> \mathbb{E}_0 \left[\int_0^t e^{-\rho u} U(C^*_{0,u}, C^*_{0,u} + G_u)\, du + \mathbb{E}_t \int_t^\infty e^{-\rho u} U(C^*_{0,u}, C^*_{0,u} + G_u)\, du \right] \\
&= \mathbb{E}_0 \int_0^\infty e^{-\rho u} U(C^*_{0,u}, C^*_{0,u} + G_u)\, du.
\end{aligned}$$

This contradicts that $\{C^*_{0,u}\}$ is the Ramsey plan. We thus conclude $\Phi^*_t = \Phi^*_0$ and that $C^*_{t,u} = C(s_u; \Phi^*_t) = C(s_u; \Phi^*_0) = C^*_{0,u}$ for all $u \geqslant t$.

□

7 Quantitative Examples

The two quantitative examples highlight how the Ramsey planner manipulates equilibrium prices and marginal utilities to manage debt costs. A recurring theme is that the state variable $X_t = \Pi_t\, U_{C,t}$ remains a continuous function of time even when the exogenous term debt process or the government expenditure process jumps discontinuously across regimes.

To illustrate how our model works, we posit a regime-switching model with two states $s_t = \{H, L\}$. We endow our representative agent with the instantaneous utility function $U(C,N) = \log C - \eta \frac{N^\gamma}{\gamma}$, where $\eta > 0$ and $\gamma \geqslant 1$, the same one employed by Debortoli,

Nunes, and Yared (2021). In subsection 7.1, we study a setting with a stochastic $b_{0-,t}$, while in subsection 7.2, we study a setting with a stochastic G_t. In both subsections, we set $\rho = 0.02$, $\eta = 1$, the transition rate λ_{HL} from H to L to 0.2 and the transition rate λ_{LH} from L to H to 0.3.

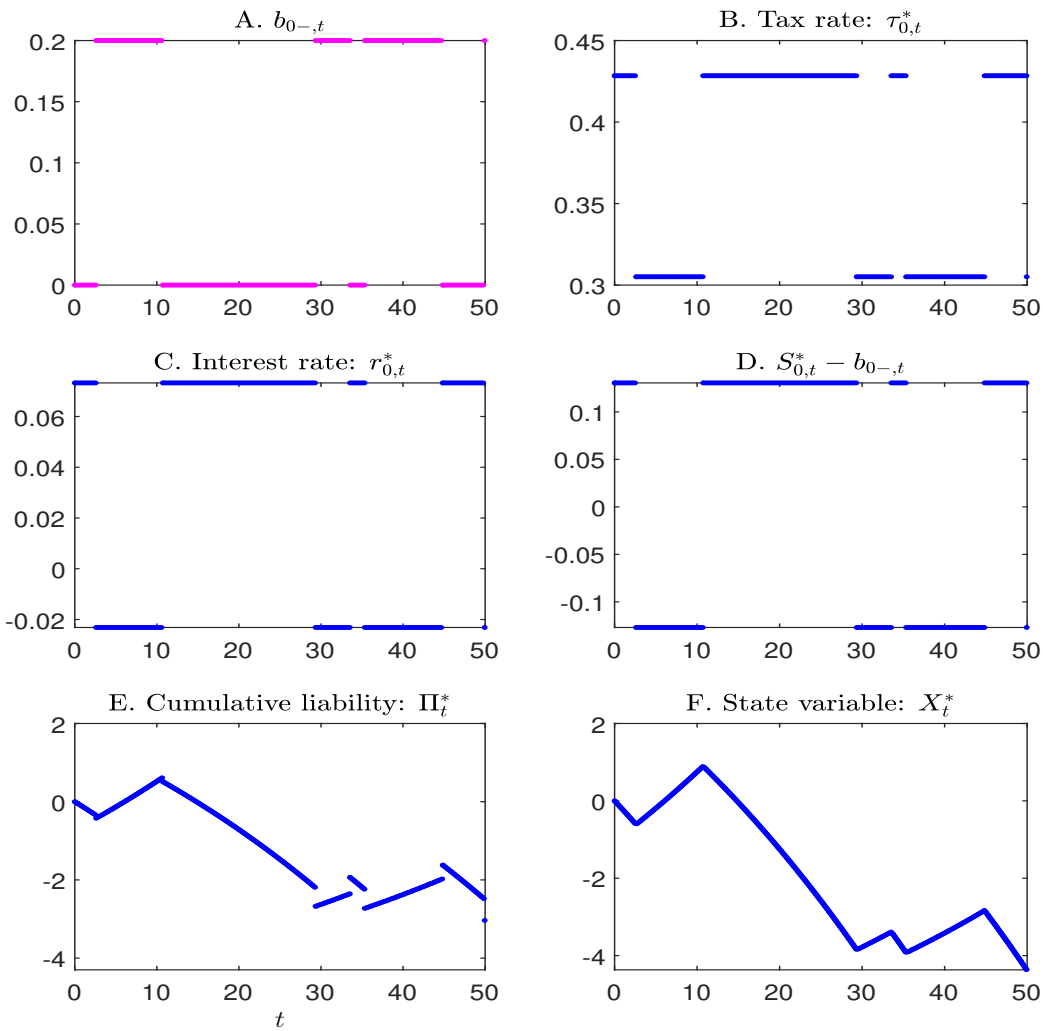

Figure 1: Ramsey plan in a two-state setting with high and low levels of initial term debt. Parameter values: $b_H = 0.2, b_L = 0, G_t = 0.2$ for $t \geqslant 0$.

7.1 State-Dependent Initial Term Debt $\{b_{0-,t}\}$

Consider a setting where the initial debt is high or low: $b_{0-,t} = b_H$ if $s_t = H$ and $b_{0-,t} = b_L$ if $s_t = L$ for $t \geqslant 0$. We set $b_H = 0.2$, $b_L = 0$, and $G_t = 0.2$ for all t. Figure 1 plots outcomes under the Ramsey plan. Panel A plots a path of $b_{0-,t}$.

Panel B shows that the Ramsey tax rate is low (high) when debt is high (low). Specifically, the tax rate $\tau^*_{0,t} = 0.31$ in state H and $\tau^*_{0,t} = 0.43$ in state L. This is because the planner wants to reduce the market value of its debt by making marginal utility in the high-debt state low. As $C^*_{0,t} = 1 - \tau^*_{0,t}$, low marginal utility, which maps to high consumption, is achieved with a low tax rate. This explains why taxes are low in high-debt states.

Panel C shows that the equilibrium interest rate is low when debt is high. Note that the interest rate $r^*_{0,t}$ is low and equals -0.023 in the high-debt state, but is high and equals 0.073 in the no-debt state. Similarly, this is because the planner wants to make the interest rate low when its debt is high in order to reduce its debt services. Panel D shows that the primary surplus $S^*_{0,t}$ in excess of the initial term debt service $b_{0-,t}$, $S^*_{0,t} - b_{0-,t}$, is low (high) when debt is high (low): $S^*_{0,t} - b_{0-,t} = 0.131$ in the high-debt state and $S^*_{0,t} - b_{0-,t} = -0.126$ in the no-debt state.[5]

Panel E shows that the implied cumulative liability Π^*_t (with no rescheduling of term debts) decreases in the no-debt state and increases in the high-debt state. Note that when the state transitions, Π^*_t generally jumps. Panel F shows that the state variable $X_t = \Pi^*_t U^*_{C,t}$, which tracks the purchasing power of Π^*_t, also decreases in the no-debt state and increases in the high-debt state. Notice that, unlike Π^*_t, X_t is continuous even when the state transitions. This is due to the household's optimal response to the government's tax plan and the associated equilibrium prices, all of which are determined at $t = 0$. The jump of Π^*_t is offset by a jump of $U^*_{C,t}$, leaving $X_t = \Pi^*_t U^*_{C,t}$ unchanged when the state transitions.

7.2 State-Dependent Government Spending $\{G_t\}$

Now government spending switches between high and low: $G_t = G_H$ if $s_t = H$ and $G_t = G_L$ if $s_t = L$ for $t \geqslant 0$. We set $G_H = 0.2, G_L = 0.15$ and $b_{0-,t} = 0.1$ for all t. Figure 2 plots outcomes under the Ramsey plan. Panel A plots a path of G_t.

[5] We obtain this result as follows. First, $C^*_{0,t} = \left(\sqrt{1 + 4b_{0-,t}\Phi^*_0(1+\Phi^*_0)} + 1\right)/(2(1+\Phi^*_0))$, which increases in $b_{0-,t}$. Second, note that $S^*_{0,t} - b_{0-,t} = \tau^*_{0,t}N^*_{0,t} - b_{0-,t} = C^*_{0,t}(1 - C^*_{0,t} - G_t) - b_{0-,t}$. Third, using $\frac{\partial C^*_{0,t}}{\partial b_{0-,t}} = \frac{\Phi^*_0}{2C^*_{0,t}(1+\Phi^*_0)-1} > 0$, we conclude that $\frac{\partial(S^*_{0,t} - b_{0-,t})}{\partial b_{0-,t}} = \frac{\partial C^*_{0,t}}{\partial b_{0-,t}}\left(2C^*_{0,t}\Phi^*_0 - G_t\right) - \Phi^*_0 - 1 < 0$.

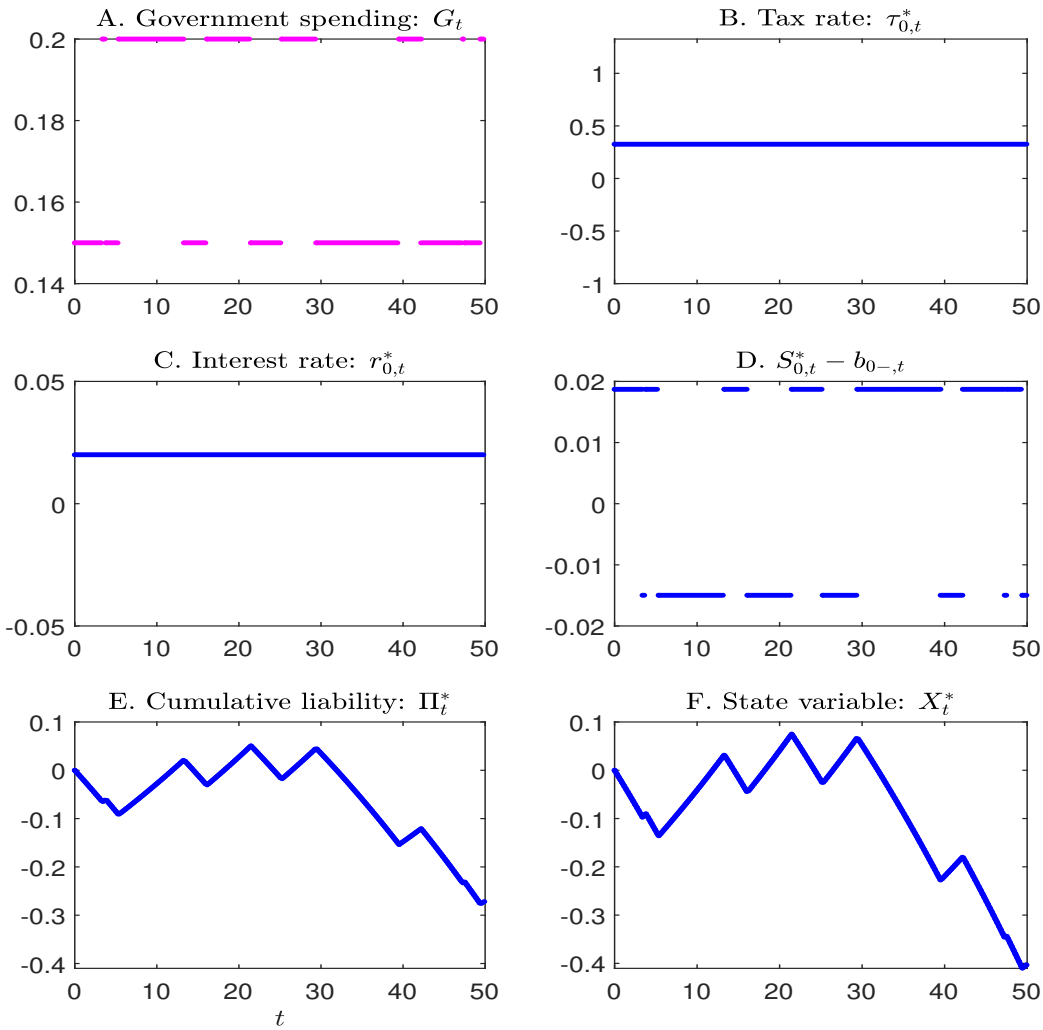

Figure 2: Ramsey plan in a two-state setting with high and low levels of government spending. Parameter values: $b_{0-,t} = 0.1$ for $t \geqslant 0$, $G_H = 0.2$, and $G_L = 0.15$.

Panel B shows that the tax rate is constant: $\tau^*_{0,t} = 1 - C^*_{0,t} = 0.33$ for all t. This is because the maximand in (28) yields an optimal C that is independent of G_t when $b_{0-,t}$ is constant. This tax-smoothing result echoes Barro (1979)'s finding that intertemporal properties of an optimal tax rate are disconnected from intertemporal properties of the government expenditure process. Since consumption $C^*_{0,t}$ is constant, the equilibrium interest rate equals the household's discount rate at all t: $r^*_{0,t} = \rho = 2\%$ (see panel C). Panel D confirms that $S^*_{0,t} - b_{0-,t}$ moves one-for-one in the opposite direction from gov-

ernment spending G_t, as both the debt service $b_{0-,t}$ and the tax rate $\tau^*_{0,t}$ are constant over time.

Unlike Figure 1, Panel E shows that the government's cumulative liability is *continuous* even as the state transitions. This follows from the tax (and consumption) smoothing in this setting. Accruing $-(S^*_{0,u} - b_{0-,u})$ from 0 to t gives $\Pi^*_t = -\int_0^t e^{\rho(t-u)}(S^*_{0,u} - b_{0-,u})\,du$. Thus, Π^*_t increases (decreases) when $S^*_{0,u} - b_{0-,u}$ is negative (positive). As $U^*_{C,t} = 1/C^*_{0,t} = 1/0.67$ for all t, $X_t = \Pi^*_t U^*_{C,t} = 1.48\Pi^*_t$, thus making Panel F proportionally track Panel E at all t.

8 Alternative Recursive Formulation

This section replaces the backward-looking cumulative liability Π_t with a forward-looking promised liability Ω_t and constructs an alternative state variable $Y_t = \Omega_t U_{C,t}$. The law of motion for Y_t contains a coefficient on a martingale increment that is indeterminate, reflecting the government's freedom in choosing Arrow securities. Despite this indeterminacy, the value function retains the same additively separable structure and the Ramsey plan is unique: the optimal consumption and tax paths are the same as those found in section 4.

We proceed in three steps. 1. For a given consumption process, we define a process $\{\Omega_t;\, t \geqslant 0\}$ that tracks the government's promised (present discounted) value of all future liabilities in the absence of any debt restructuring. 2. From the $\{\Omega_t;\, t \geqslant 0\}$ process, we construct an alternative state variable process $\{Y_t;\, t \geqslant 0\}$. 3. We frame a dynamic program with $\{Y_t;\, t \geqslant 0\}$ as a state variable and a state-s_t-dependent value function, $K(Y_t, s_t)$, that is additively separable: $K(Y_t, s_t) = -\Psi_0 Y_t + H(s_t;\, \Psi_0)$, where Ψ_0 is a scalar to be determined.

In addition to providing an alternative approach for solving the Ramsey plan, we also discuss difficulties in implementing the Ramsey plan if Arrow securities are not chosen properly.

8.1 Cumulative Liabilities with no Debt Restructuring

For a given competitive equilibrium (see Definition 1), we can construct the following $\{\Omega_t\}$ process:

$$\Omega_t = \mathbb{E}_t \int_t^\infty \frac{q_{0,u}}{q_{0,t}} \left(b_{0-,u} + G_u - \tau_{0,u} N_{0,u}\right) du\,, \quad \Omega_0 = 0\,, \tag{53}$$

where the ratio $q_{0,u}/q_{0,t}$ discounts the flow $(b_{0-,u} + G_u - \tau_{0,u}\, N_{0,u})\, du$ back to the government's time-t promised liability. This $\{\Omega_t\}$ process is the government's time-t discounted value of

its liabilities in the absence of debt restructuring from t to ∞.

Note that

$$q_{0,t}\Pi_t + q_{0,t}\Omega_t = \mathbb{E}_t\left[\int_0^\infty q_{0,u}\left(b_{0-,u} + G_u - \tau_{0,u}N_{0,u}\right)du\right] \tag{54}$$

is a martingale. Using the martingale representation theorem, we obtain:

Lemma 3. *For a given $\{\alpha_t; t \geqslant 0\}$ process set at $t = 0$, we can express the dynamics for the $\{\Omega_t; t \geqslant 0\}$ process defined in (53) as follows:*

$$\begin{aligned} d\Omega_t \quad &= \quad r_{0,t-}\Omega_{t-}dt + \left(b_{0-,t-} + G_{t-} - \tau_{0,t-}N_{0,t-}\right)dt \\ &\quad + \sum_{j\neq s_{t-}} \alpha_{t-}\Omega_{t-}\Big[\underbrace{d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j}e^{\eta(s_{t-},j)}dt}_{\textit{increment of a martingale}}\Big], \end{aligned} \tag{55}$$

where $\alpha_{t-} = \alpha(s_{t-}, j)$ for $j \neq s_{t-}$ and $\eta(s_{t-}, s_t) = \ln\frac{U_{C,t}}{U_{C,t-}}$.

It is important to note that the $\{\Omega_t\}$ process defined in (53) is conceptually different from the $\{\Pi_t\}$ process defined in (18), which tracks cumulative liabilities accrued from 0 to t in the complete absence of debt restructuring. Comparing their respective dynamics, (55) and (19), we see that the only difference is the last term that involves the increment of a martingale, which captures the state transition. For the cumulative liability Π_t process, the coefficient multiplying the martingale increment in (19) is unique and equals $\left(e^{-\eta(s_{t-},j)} - 1\right)\Pi_{t-}$. By contrast, for the promised liability Ω_t process, the coefficient multiplying the martingale increment in (55), $\alpha_{t-}\Omega_{t-}$, is indeterminate. This is because the cumulative liability $\{\Pi_t\}$ process is "backward-looking," while the promised liability $\{\Omega_t\}$ process is "forward-looking." That is, there exists a unique $\alpha(s_t, j)$ process, namely, $\alpha(s_t, j) = \left(e^{-\eta(s_t,j)} - 1\right)$, that makes $\Pi_t = \Omega_t$ for all $t \geqslant 0$.

Next, we use the $\{\Omega_t\}$ process to introduce a state variable $\{Y_t; t \geqslant 0\}$.

8.2 A State Variable Y_t

Inspired by the martingale property of the $\{q_{0,t}\Pi_t + q_{0,t}\Omega_t\}$ process, we define

$$Y_t = \Omega_t U_{C,t}\,, \tag{56}$$

which will serve as a state variable for a recursive formulation of the Ramsey problem. Evidently, Y_t incorporates information about both Ω_t and the marginal utility of consumption

$U_{C,t}$.

Combining (56) and (5) yields

$$Y_t = \Omega_t e^{\rho t} q_{0,t} U_{C,0}\,. \tag{57}$$

Differentiating both sides of the above equation and using $C_t = N_t - G_t$, $e^{\rho t} U_{C,0} q_{0,t} = U_{C,t}$, and $(1-\tau_{0,t})U_{C,t} = -U_{N,t}$, as implied by (1) and (4)-(5), we obtain

$$\begin{aligned} dY_t &= (\rho Y_{t-} + b_{0-,t-} U_{C,t-} - C_{0,t-} U_{C,t-} - N_{0,t-} U_{N,t-})\, dt \\ &\quad + \sum_{j \neq s_{t-}} \beta_{t-} Y_{t-} \big(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} e^{\eta(s_{t-},j)} dt \big)\,, \end{aligned} \tag{58}$$

where

$$\beta_{t-} = \beta(s_{t-}, j) = \big[1 + \alpha(s_{t-}, j)\big] e^{\eta(s_{t-},j)} - 1\,. \tag{59}$$

We have established:

Proposition 7. *For a Ramsey plan that commits to $\{\alpha_t; t \geqslant 0\}$, Y_t defined in (56) evolves according to (58).*

Note that, as we discussed earlier, if and only if $\alpha(s_{t-}, j) = e^{-\eta(s_{t-},j)} - 1$ for all t, the promised liability process $\{\Omega_t\}$ coincides with the cumulative liability process $\{\Pi_t\}$ at all t, which in turn implies that the $\{Y_t;\, t \geqslant 0\}$ process defined in (56) coincides with the $\{X_t; t \geqslant 0\}$ process defined in (20).

In section 8.3, we cast the Ramsey problem in terms of the $\{Y_t; t \geqslant 0\}$ process in two steps: 1) for a given Y_0, we define and solve a dynamic programming (DP) problem with Y_t and s_t as state variables; 2) we characterize the Ramsey plan.

8.3 Dynamic Program (DP) and Ramsey Plan

Definition 3 (DP Problem). *Confronting Y_0 at $t = 0$, the government solves* (23) *subject to the law of motion (58).*

Let $K(Y_0, s_0)$ denote the (optimal) value function for this problem. To solve $K(Y_0, s_0)$, we first construct the following time$-t$ optimization problem:

$$\max_{\{C_{0,u},\, \beta_u; u \geqslant t\}} \mathbb{E}_t \left[\int_t^\infty e^{-\rho(u-t)} U(C_{0,u}, C_{0,u} + G_u) du \right] \tag{60}$$

subject to (58). Let $K(Y_t, s_t)$ denote the value function where (Y_t, s_t) are the state variables. In Appendix A, we show that $K(Y_t, s_t)$ satisfies the following HJB equation:

$$\rho K(Y_t, s_t) = \max_{C_{0,t},\, \beta_t} \; U(C_{0,t}, C_{0,t} + G_t) + \frac{\partial K}{\partial Y_t}\left[\rho Y_t - C_{0,t} U_{C,t} - (C_{0,t} + G_t) U_{N,t} + b_{0-,t} U_{C,t}\right] + \sum_{j \neq s_t} \lambda_{s_t,j} \left[K(Y + \beta_t Y, j) - K(Y, s_t) - \beta_t \frac{\partial K}{\partial Y_t}\right]. \tag{61}$$

We guess and verify that $K(Y_t, s_t)$ takes the following form:

$$K(Y_t, s_t) = -\Psi_0 Y_t + H(s_t; \Psi_0)\,, \tag{62}$$

where Ψ_0 is an object to be chosen by the Ramsey planner and $H(s_t; \Psi_0)$ is given by (27). Note that the second term in (62) has the same functional form as the second term in (26): $H(s_t; \Psi_0)$ in (62) is also characterized by Proposition 4.

The $\{\beta_t; t \geqslant 0\}$ process associated with the $\{Y_t; t \geqslant 0\}$ process given in (56) is indeterminate, a consequence of the additively separable form (62) of the value function $K(Y_t, s_t)$.

Replacing Φ_0 in the optimal consumption and labor supply expressions (28) and (29) with Ψ_0, we obtain the optimal consumption $C(s_t; \Psi_0)$ and labor supply $N(s_t; \Psi_0)$ for this alternative recursive formulation, paralleling the analysis in subsection 4.3.

In general, for a competitive equilibrium, Ψ_0 in (62) does not equal Φ_0 in (26). But under the Ramsey plan, $\Psi_0^* = \Phi_0^*$, which we show below.

The implementability constraint (7) requires that Ψ_0 satisfy $I_0(s_0; \Psi_0) = 0$, where $I_0(\cdot\,;\,\cdot)$ is defined in (32). Thus Ψ_0 has the same admissible set as Φ_0, i.e., $\Psi_0 \in \mathcal{A}$, where $\mathcal{A}$ is defined in subsection 4.4. Using (62) and $Y_0 = 0$, the Ramsey planner chooses Ψ_0 by solving:

$$M(s_0) \;=\; \max_{\Psi_0 \in \mathcal{A}} K(0, s_0; \Psi_0) \;=\; \max_{\Psi_0 \in \mathcal{A}} H(s_0; \Psi_0)\,. \tag{63}$$

Because both problems reduce to maximizing $H(s_0; \cdot)$ over the same admissible set $\mathcal{A}$, problem (63) coincides with problem (33). It follows that $\Psi_0^* = \Phi_0^*$, so the optimal consumption policy $C(s_t; \Psi_0^*)$ coincides with $C(s_t; \Phi_0^*)$ for all s_t and $\{C(s_t; \Psi_0^*); t \geqslant 0\} = \{C(s_t; \Phi_0^*); t \geqslant 0\}$. Therefore, we have established:

Proposition 8. *When a Ramsey plan exists, the problem* (63) *is equivalent to the problem* (33)*, and* $\Psi_0^* = \Phi_0^*$ *and* $M(s_0) = W(s_0)$ *hold.*

9 Concluding Remarks

This paper extended the deterministic continuous-time formulation of the Lucas-Stokey optimal tax problem in Jiang et al. (2024) to a stochastic environment driven by a Markov chain. Several lessons emerged.

Instantaneous debt as a sufficient instrument. A central message of both the deterministic predecessor and this stochastic extension is that the Ramsey plan can be implemented without rescheduling the initial term structure of government debt. It suffices for the government to trade locally risk-free instantaneous debt — together with, in the stochastic setting, competitively priced hedging contracts — while leaving the inherited coupon stream $\{b_{0-,t}\}$ untouched. This contrasts with Lucas and Stokey (1983), who implemented the Ramsey plan by restructuring the entire term structure period by period.

Complete markets in continuous time. Translating Lucas and Stokey's (1983) complete-markets assumption to continuous time required care. The dynamic spanning constructions pioneered by Black and Scholes (1973), Merton (1973), and Harrison and Kreps (1979) show that instantaneous debt together with $n-1$ hedging contracts that pay off at Markov state transitions suffice to span all contingencies over each infinitesimal interval. These hedging contracts allow the government to offset jumps in bond prices and in its cumulative liability at state transitions, maintaining dynamic completeness without access to a full set of dated Arrow securities. $n-1$ is not defined. to discuss.

A natural state variable. The cumulative government liability Π_t — the running present-value sum of primary deficits and inherited debt service — can jump at state transitions because bond prices and tax revenues jump. Yet its purchasing-power-adjusted counterpart $X_t = \Pi_t U_{C,t}$ evolves continuously in time. This continuity makes X_t a tractable state variable and underlies the additively separable structure $V(X_t, s_t) = -\Phi_0 X_t + H(s_t; \Phi_0)$ of the Ramsey planner's value function. The separability in turn enables us to reduce the planner's infinite-dimensional problem to a one-dimensional search over the scalar Φ_0.

Local commitment and sequential implementation. Our timing protocol grants each government only "local commitment": its current-period tax rate is set by its immediate predecessor. Under this protocol, the continuation government at date t inherits

instantaneous debt $B_t = \Pi_t^*$ and the tail of the original term debt, and faces the same purchasing-power-adjusted state X_t^* as the date-0 Ramsey planner. Because the value function is linear in X_t, the continuation government's incentives are aligned with those of its predecessor, and it voluntarily confirms the Ramsey plan. This structure helps us understand and address the Debortoli, Nunes, and Yared (2021) counterexample.

Tax smoothing and hedging. The quantitative illustrations in section 7 revealed a sharp dichotomy. When only the initial term debt switches between regimes, the Ramsey planner manipulates interest rates and bond prices to reduce the market value of outstanding obligations, allowing taxes and consumption to vary across states. When instead only government spending switches while the initial term debt is constant, the planner achieves perfect tax smoothing: consumption, tax rates, and interest rates all remain constant across states, funded entirely by adjusting instantaneous debt and hedging positions. The latter result extends to continuous time Barro's (1979) insight that optimal tax rates need not inherit the serial correlation properties of the government expenditure process.

Uniqueness despite indeterminacy. Section 8 showed that an alternative recursive formulation built on a forward-looking promised liability Ω_t leads to indeterminacy in the portfolio process — the government has freedom in how it distributes obligations across hedging contracts. Nevertheless, the Ramsey plan itself is unique: the optimal multiplier Ψ_0^* equals Φ_0^*, and consumption-tax paths coincide with those from the backward-looking formulation in section 4. Indeterminacy is confined to the portfolio that finances a unique allocation.

References

Arrow, K. J. (1964). The role of securities in the optimal allocation of risk-bearing. *The review of economic studies 31*(2), 91–96.

Barro, R. J. (1979). On the determination of the public debt. *Journal of political Economy 87*(5, Part 1), 940–971.

Black, F. and M. Scholes (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 637–654.

Debortoli, D., R. Nunes, and P. Yared (2021). Optimal fiscal policy without commitment: Revisiting lucas-stokey. *Journal of Political Economy 129*(5), 1640–1665.

Duffie, D. (2001). *Dynamic asset pricing theory*. Princeton University Press.

Harrison, J. M. and D. M. Kreps (1979). Martingales and arbitrage in multiperiod securities markets. *Journal of Economic theory 20*(3), 381–408.

Jiang, W., T. J. Sargent, and N. Wang (2024, August). A dynamic program and debt management policy for implementing a ramsey tax plan. Working paper.

Lucas, R. E. J. and N. L. Stokey (1982, 8). Optimal fiscal and monetary policy in an economy without capital. Discussion Paper 532, Northwestern University, Kellogg School of Management, Center for Mathematical Studies in Economics and Management Science, Evanston, IL.

Lucas, R. E. J. and N. L. Stokey (1983). Optimal fiscal and monetary policy in an economy without capital. *Journal of Monetary Economics 12*(1), 55–93.

Merton, R. C. (1973). Theory of rational option pricing. *The Bell Journal of Economics and Management Science 4*(1).

Appendices

A Technical Details

Regularity Conditions for Proposition 2. We provide two regularity conditions that guarantee that a Ramsey plan exists. First, for a given initial debt structure, the admissible allocation set is not empty, so there exist allocations $\{C_{0,t};\, t \geqslant 0\}$ that satisfy the implementability condition (7). Second, $(CU_C + (C+G)U_N - bU_C)$ is concave for $G > 0$, $b \geqslant 0$.

Proof of Lemma 1: Dynamics of the $\{\Pi_t\}$ Process. Below, we prove Lemma 1 by deriving the dynamics of Π_t defined in (18). Using the property that $\{X_t\}$ given in (22) is continuous in t and combining (20) with (5), we obtain

$$\Pi_t = q_{0,t}^{-1} \int_0^t q_{0,u} \left(b_{0-,u} + G_u - \tau_{0,u} N_{0,u}\right) du\,. \tag{A-1}$$

Given a consumption process $\{C_{0,t}; t \geqslant 0\}$, the SDF can be written as (see, e.g., Duffie (2001)):

$$dq_{0,t} = -r_{0,t-} q_{0,t-} dt - q_{0,t-} \sum_{j \neq s_{t-}} \zeta(s_{t-}, j) \left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j} dt \right)\,, \tag{A-2}$$

where $\zeta(s_{t-}, j)$ is given by

$$\zeta(i,j) = -\left(e^{\eta(i,j)} - 1\right)\,, \tag{A-3}$$

and $\eta(i,j)$ for $i \neq j$ is given by:

$$\eta(i,j) = \ln \frac{U_C(C(j), N(j))}{U_C(C(i), N(i))} \tag{A-4}$$

Applying Ito's lemma to (A-1) and using (22) and (A-2), we obtain

$$\begin{aligned} d\Pi_t =& q_{0,t-}^{-1} d\left[\int_0^t q_{0,u} \left(b_{0-,u} + G_u - \tau_{0,u} N_{0,u}\right) du\right] + \left[\int_0^{t-} q_{0,u} \left(b_{0-,u} + G_u - \tau_{0,u} N_{0,u}\right) du\right] d\left(q_{0,t}^{-1}\right) \\ &+ (\Pi_t - \Pi_{t-}) \sum_{j \neq s_{t-}} d\mathcal{J}_t^{(s_{t-},j)} \\ =& \left(b_{0-,t-} + G_{t-} - \tau_{0,t-} N_{0,t-}\right) dt \end{aligned}$$

$$
\begin{aligned}
&- \left[\int_0^{t-} q_{0,u}\left(b_{0-,u} + G_u - \tau_{0,u}N_{0,u}\right)du\right] q_{0,t-}^{-2}\big(-r_{0,t-}q_{0,t-}dt + q_{0,t-}\sum_{j\neq s_{t-}}\zeta(s_{t-},j)\lambda_{s_{t-},j}dt\big) \\
&\quad + (\Pi_t - \Pi_{t-})\sum_{j\neq s_{t-}} d\mathcal{J}_t^{(s_{t-},j)} \\
&= \left(b_{0-,t-} + G_{t-} - \tau_{0,t-}N_{0,t-}\right)dt - \Pi_{t-}\left(-r_{0,t-}dt + \sum_{j\neq s_{t-}}\zeta(s_{t-},j)\lambda_{s_{t-},j}dt\right) \\
&\quad + (\Pi_t - \Pi_{t-})\sum_{j\neq s_{t-}} d\mathcal{J}_t^{(s_{t-},j)} \\
&= \left(b_{0-,t-} + G_{t-} - \tau_{0,t-}N_{0,t-}\right)dt + r_{0,t-}\Pi_{t-}dt \\
&\quad + \Pi_{t-}\sum_{j\neq s_{t-}}\left[\left(\frac{\Pi_t}{\Pi_{t-}} - 1\right)d\mathcal{J}_t^{(s_{t-},j)} - \zeta(s_{t-},j)\lambda_{s_{t-},j}dt\right] \\
&= \left(b_{0-,t-} + G_{t-} - \tau_{0,t-}N_{0,t-}\right)dt + r_{0,t-}\Pi_{t-}dt \\
&\quad + \sum_{j\neq s_{t-}}\left(e^{-\eta(s_{t-},j)} - 1\right)\Pi_{t-}\left(d\mathcal{J}_t^{(s_{t-},j)} - \lambda_{s_{t-},j}e^{\eta(s_{t-},j)}dt\right) \qquad \text{(A-5)}
\end{aligned}
$$

where the last equality holds because that $\left(\frac{\Pi_t}{\Pi_{t-}} - 1\right) = e^{-\eta(s_{t-},s_t)} - 1$ and $\zeta(s_{t-},s_t) = -\left(e^{\eta(s_{t-},s_t)} - 1\right) = e^{\eta(s_{t-},s_t)}\left(e^{-\eta(s_{t-},s_t)} - 1\right)$.

Derivation of HJB Equation (25)**.** Next, we construct a martingale that helps characterize the process

$$\int_0^t e^{-\rho u}U(C_{0,u}, C_{0,u} + G_u)\,du + e^{-\rho t}V(X_t, s_t).$$

Under the physical measure, this process is a martingale, and hence its drift must be zero. Therefore,

$$\mathbb{E}_t\left[d\left(e^{-\rho t}V(X_t,s_t)\right)\right] + e^{-\rho t}U(C_{0,t}, C_{0,t} + G_t)\,dt = 0. \qquad \text{(A-6)}$$

Applying Ito's lemma to the left hand side of (A-6), we obtain the HJB equation (25).

Proof of Proposition 4 Substituting optimal consumption defined in (28) into HJB equation (27), we oabtain for $s \in (1, 2, \cdots, n)$

$$\rho H(s;\Phi_0) = A(s;\Phi_0) + \sum_{k\neq s}\lambda_{s,k}\left[H(k;\Phi_0) - H(s;\Phi_0)\right]. \qquad \text{(A-7)}$$

We then obtain

$$(\rho\mathbb{I} - \Lambda)\,\mathbb{H} = \mathbb{A}\,, \tag{A-8}$$

which gives (30).